Devopslab project 7

```
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Restarting existing docker container for "minikube" ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, metrics-server, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\aaroh>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\aaroh>mkdir project7-mongo

C:\Users\aaroh>cd project7-mongo

C:\Users\aaroh\project7-mongo>cd
C:\Users\aaroh\project7-mongo

C:\Users\aaroh\project7-mongo>notepad mongo-secret.yaml

C:\Users\aaroh\project7-mongo>
```

mongo-secret.yaml

File Edit View

```yaml
apiVersion: v1
kind: Secret
metadata:
  name: mongo-secret
type: Opaque
stringData:
  mongo-user: admin
  mongo-password: password123
```

Ln 8, Col 30 | 133 characters | Plain text | 100% | Windows (CR | UTF-8

```
C:\Users\aaroh\project7-mongo>notepad mongo-secret.yaml

C:\Users\aaroh\project7-mongo>kubectl apply -f mongo-secret.yaml
secret/mongo-secret created

C:\Users\aaroh\project7-mongo>kubectl get secrets
NAME                                                                  TYPE                 DATA   AGE
alertmanager-monitoring-kube-prometheus-alertmanager                  Opaque               1      39d
alertmanager-monitoring-kube-prometheus-alertmanager-cluster-tls-config   Opaque           1      39d
alertmanager-monitoring-kube-prometheus-alertmanager-generated        Opaque               1      39d
alertmanager-monitoring-kube-prometheus-alertmanager-tls-assets-0     Opaque               0      39d
alertmanager-monitoring-kube-prometheus-alertmanager-web-config       Opaque               1      39d
loki                                                                  Opaque               1      39d
loki-promtail                                                         Opaque               1      39d
mongo-secret                                                          Opaque               2      7s
monitoring-grafana                                                    Opaque               3      39d
monitoring-kube-prometheus-admission                                  Opaque               3      39d
prometheus-monitoring-kube-prometheus-prometheus                      Opaque               1      39d
prometheus-monitoring-kube-prometheus-prometheus-thanos-prometheus-http-client-file   Opaque   1   39d
prometheus-monitoring-kube-prometheus-prometheus-tls-assets-0         Opaque               1      39d
prometheus-monitoring-kube-prometheus-prometheus-web-config           Opaque               1      39d
sh.helm.release.v1.loki.v1                                            helm.sh/release.v1   1      39d
sh.helm.release.v1.monitoring.v1                                      helm.sh/release.v1   1      39d

C:\Users\aaroh\project7-mongo>
```

mongo-secret.yaml | mongo-configmap.yaml

File Edit View

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: mongo-config
data:
  mongo-host: mongo-service
```

```
C:\Users\aaroh\project7-mongo>notepad mongo-configmap.yaml

C:\Users\aaroh\project7-mongo>kubectl apply -f mongo-configmap.yaml
configmap/mongo-config created

C:\Users\aaroh\project7-mongo>kubectl get configmaps
NAME                                                      DATA   AGE
kube-root-ca.crt                                          1      39d
loki-loki-stack                                           1      39d
loki-loki-stack-test                                      1      39d
mongo-config                                              1      7s
monitoring-grafana                                        1      39d
monitoring-grafana-config-dashboards                      1      39d
monitoring-kube-prometheus-alertmanager-overview          1      39d
monitoring-kube-prometheus-apiserver                      1      39d
monitoring-kube-prometheus-cluster-total                  1      39d
monitoring-kube-prometheus-controller-manager             1      39d
monitoring-kube-prometheus-etcd                           1      39d
monitoring-kube-prometheus-grafana-datasource             1      39d
monitoring-kube-prometheus-grafana-overview               1      39d
monitoring-kube-prometheus-k8s-coredns                    1      39d
monitoring-kube-prometheus-k8s-resources-cluster          1      39d
monitoring-kube-prometheus-k8s-resources-multicluster     1      39d
monitoring-kube-prometheus-k8s-resources-namespace        1      39d
monitoring-kube-prometheus-k8s-resources-node             1      39d
monitoring-kube-prometheus-k8s-resources-nodes-overview   1      39d
monitoring-kube-prometheus-k8s-resources-pod              1      39d
```

mongo-secret.yaml | mongo-configmap.yaml | mongo-deployment.yaml

File Edit View

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongo
  template:
    metadata:
      labels:
        app: mongo
    spec:
      containers:
        - name: mongodb
          image: mongo:latest
          ports:
            - containerPort: 27017
          env:
            - name: MONGO_INITDB_ROOT_USERNAME
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-user

            - name: MONGO_INITDB_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-password
```

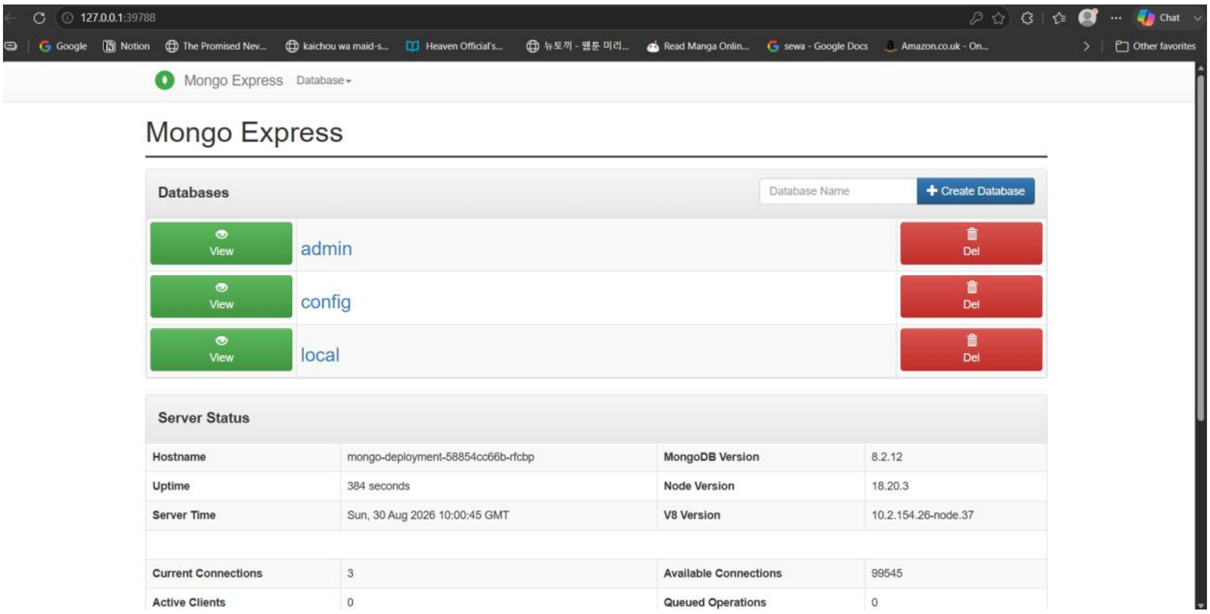
```
C:\Users\aaroh\project7-mongo>notepad mongo-deployment.yaml

C:\Users\aaroh\project7-mongo>kubectl apply -f mongo-deployment.yaml
deployment.apps/mongo-deployment created

C:\Users\aaroh\project7-mongo>kubectl get deployments
NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
mongo-deployment                      0/1     1            0           5s
monitoring-grafana                    0/1     1            0           39d
monitoring-kube-prometheus-operator   1/1     1            1           39d
monitoring-kube-state-metrics         1/1     1            1           39d
nginx                                 1/1     1            1           39d
social-media-app                      2/2     2            2           13d

C:\Users\aaroh\project7-mongo>
```

mongo-secret.yaml | mongo-configmap.yaml | mongo-deployment.yaml | mongo-service.yaml | mongo-express-deployment.yaml

File Edit View

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo-express-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongo-express
  template:
    metadata:
      labels:
        app: mongo-express
    spec:
      containers:
        - name: mongo-express
          image: mongo-express:latest
          ports:
            - containerPort: 8081
          env:
            - name: ME_CONFIG_MONGODB_SERVER
              valueFrom:
                configMapKeyRef:
                  name: mongo-config
                  key: mongo-host

            - name: ME_CONFIG_MONGODB_PORT
              value: "27017"

            - name: ME_CONFIG_MONGODB_ADMINUSERNAME
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-user

            - name: ME_CONFIG_MONGODB_ADMINPASSWORD
              valueFrom:
                secretKeyRef:
                  name: mongo-secret
                  key: mongo-password
```

```
C:\Users\aaroh\project7-mongo>notepad mongo-express-deployment.yaml

C:\Users\aaroh\project7-mongo>kubectl apply -f mongo-express-deployment.yaml
deployment.apps/mongo-express-deployment created

C:\Users\aaroh\project7-mongo>kubectl get pods
NAME                                                   READY   STATUS              RESTARTS        AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0 2/2     Running             10 (13d ago)    39d
load-generator                                         0/1     Error               0               13d
loki-0                                                 1/1     Running             7 (8m53s ago)   39d
loki-promtail-7c7t4                                    1/1     Running             5 (13d ago)     39d
mongo-deployment-58854cc66b-rfcbp                      1/1     Running             0               5m26s
mongo-express-deployment-65df65bf54-tv6pt              0/1     ContainerCreating   0               6s
monitoring-grafana-547fcbbdc9-hpr22                    2/3     CrashLoopBackOff    60 (2m55s ago)  39d
monitoring-kube-prometheus-operator-8876fbff5-6jnp2    1/1     Running             8 (9m36s ago)   39d
monitoring-kube-state-metrics-6cbbf66fff-qj6tn         1/1     Running             9               39d
monitoring-prometheus-node-exporter-hbrbk              1/1     Running             5 (13d ago)     39d
nginx-56c45fd5ff-87q74                                 1/1     Running             6 (13d ago)     39d
prometheus-monitoring-kube-prometheus-prometheus-0     2/2     Running             10 (13d ago)    39d
social-media-app-c859d545-hnxtj                        1/1     Running             1 (13d ago)     13d

C:\Users\aaroh\project7-mongo>
```




```
C:\Users\aaroh>kubectl get configmap mongo-config
NAME           DATA   AGE
mongo-config   1      10m

C:\Users\aaroh>kubectl get secret mongo-secret
NAME           TYPE     DATA   AGE
mongo-secret   Opaque   2      12m

C:\Users\aaroh>kubectl get all -l app=mongo
NAME                                    READY   STATUS    RESTARTS   AGE
pod/mongo-deployment-58854cc66b-rfcbp   1/1     Running   0          9m44s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/mongo-deployment-58854cc66b   1         1         1       9m44s

C:\Users\aaroh>
```

Conclusion

Successfully deployed **MongoDB and Mongo Express using Kubernetes and Minikube**, with ConfigMap and Secret for configuration and credentials. Verified that Mongo Express successfully connected to MongoDB and displayed the databases.